

CT500 FirmWare Programming Guide

Version 1.0

Shmt Co., Ltd.

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Technical documents for Firmware

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1. INTRODUCTION

본 문서는 CT500 의 firmware programming 시 사용 가능한 라이브러리 함수들에 대한내용을 다룬다.

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2. DRIVERS FUNCTIONS

CT500 의 driver 함수는 pre/post-level driver 함수로 구성된다. Pre-level 드라이버 함수는 각 하드웨어 모듈의 레지스터 설정과 관련된 함수들이며, post-level 드라이버 함수들은 pre-level 드라이버를 이용한 각 하드웨어 모듈의 read/write와 같은 functional한 동작을 위한 드라이버 함수들이다.

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3. BASIC PERIPHERALS

3.1. DMA

3.1.1. DMA2Init()

Description

DMA의 Schedule 및 Resource를 초기화 한다.

시스템 시작과 동시에 한번만 실행을 해주면 된다.

```
void DMA2Init( smtBoolean polling)
```

Parameters

[Polling](#)

DMA동작 완료를 Polling으로 처리할지 인터럽트로 처리할지를 결정한다.

SMT_TRUE일 경우 Polling으로 처리하고 SMT_FALSE 일경우 인터럽트로 처리한다.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.1.2. smt2EDMACopy()

Description

Peripheral에서 memory로 DMA를 이용하여 데이터를 전송하는데 사용이 된다.

Memory에서 Memory로는 허용되지 않으며 그때는 smt2DMAMemCopy()를 사용해야 한다.

```
smtUInt32 smt2EDMACopy(DMA_DEVICE device, EDMA_STRUCT *dma, smtUInt8 descCount);
```

Parameters

device

EDMA에서는 미리정의된 peripheral에 따라 Channel이 정의되어 있다. 미리 정의된 device일 경우 해당하는 디바이스 값을 설정하고 그렇지 않을 경우 DMADEV_NONE를 사용한다.

Enumerator

Enumerator	Description
DMADEV_NAND	NAND device
DMADEV_SEIPTX / DMADEV_SEIPRX	SEIP device
DMADEV_I2STX / DMADEV_I2SRX	I2S device
DMADEV_SPI0TX / DMADEV_SPI0RX	SPI device
DMADEV_UART0TX / DMADEV_UART0RX DMADEV_UART1TX / DMADEV_UART1RX	UART 0/1 device
DMADEV_NONE	none

dma

EDMA 설정값이 있다. Src, dst를 지정하고 각각의 bus width 및 한번에 전송할 블록등을 설정한다.

```
typedef struct
{
    smtUInt32 src;           // Source address
    smtUInt32 dst;          // Destination address
    smtUInt32 nBytes;        // Transfer length

    smtBoolean m2m;         // Memory to memory transfer flag
    DMA_BUSWIDTH srcWidth;  // Source bus width
    DMA_BUSWIDTH dstWidth;  // Destination bus width

    smtUInt8 tSize;         // Transfer size

    smtBoolean srcInc;       // Source address increment flag
    smtBoolean dstInc;       // Destination address increment flag
}
```

```
smtBoolean startIntEn;    // Start interrupt enable flag
smtBoolean EndIntEn;      // End interrupt enable flag

ptDMAIntFunction intFunction; // DMA interrupt function pointer
} EDMA_STRUCT;
```

descCount

디스크립터(DMA를 이용하여 전송하려는 내용을 여러 개를 디스크립터로 만들어서 한번에 적용하는데 사용)를 사용할경우 설정한 디스크립터 개수가 들어간다.

그렇지 않을경우 1 을 넣는다.(디스크립터를 사용하지 않음)

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
EDMA_STRUCT edma;

edma.src      = (smtUInt32)&NANDDATA;
edma.dst      = (smtUInt32)ppagebuf;
edma.nBytes   = sizeInByte;

edma.m2m      = SMT_FALSE;
edma.srcWidth = transSize;
edma.dstWidth = DMA_BUSWIDTH32;
edma.tSize    = burstSize;

edma.srcInc = DMA_ADDR_NOINC;
edma.dstInc = DMA_ADDR_INC;

edma.startIntEn = SMT_FALSE;
edma.EndIntEn   = SMT_FALSE;

edma.intFunction = &NANDDMAHandler;

smt2EDMACopy(DMADEV_NAND, &descriptor, 1);
```

3.1.3. smt2GDMACopy()

Description

2D 메모리영역을 2D 메모리 영역으로 복사하는데 사용이 된다. 이미지 복사에 사용이 된다.

```
smtUInt32 smt2GDMACopy(GDMA_STRUCT *gdma);
```

Parameters

gdma

source 이미지의 크기(width, height)와 destination 이미지의 크기와 복사할 영역(sClipX, sClipY)와 복사될 영역(dClipX, dClipY)가 필요하다.

```
typedef struct
{
    smtUInt32 src;                // Source address
    smtUInt32 sWidth, sHeight;    // Source image width, height
    smtUInt32 sClipX, sClipY;     // Source start position

    smtUInt32 dst;                // Destination address
    smtUInt32 dWidth, dHeight;    // Destination image, width, height
    smtUInt32 dClipX, dClipY;     // Destination start position

    smtUInt32 clipWidth, clipHeight; // Copied image width, height

    smtUInt32 bpp;                // Pixel format

    ptDMAIntFunction intFunction; // DMA interrupt function pointer
} GDMA_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
gdma.src = (smtUInt32)src_addr;
gdma.sWidth = 16;
gdma.sHeight = 16; // source image 크기
```

```
gdma.sClipX = 2; // 복사할 영역의 시작 좌표
gdma.sClipY = 2;

gdma.dst = (smtUint32)dst_addr;
gdma.dWidth = 16;
gdma.dHeight = 16; // destination image 크기

gdma.dClipX = 4;
gdma.dClipY = 2; // 복사될 영역의 시작 좌표

gdma.clipWidth = 8;
gdma.clipHeight = 8; // 복사할 이미지 영역

gdma.bpp = 1; // Bit Per Pixel
gdma.intFunction = &GDMAHandler; // 복사 완료후 수행할 루틴.

errCode = smt2GDMACopy(&gdma);
```

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3.1.4. smt2DMAMemCopy ()

Description

DMA를 이용한 Malloc 함수.

```
smtUint32 smt2DMAMemCopy(smtUint32 *src, smtUint32 *dst, smtUint32 length)
```

Parameters

src

source 메모리 영역

dst

target 메모리 영역

length

byte 단위 복사할 Byte

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

None

3.2. GPIO

3.2.1. smtGPIOSetDefault()

Description

GPIO0/GPIO1 port (32*2) default setting

```
void smtGPIOSetDefault(void);
```

Parameters

none

Return Value

none.

Remarks

Input mode, GPIO interrupt disable, Single active low edge interrupt

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.2. smtGPIOSetDir()/smtGPIOGetDir()

Description

GPIO GPIO0_OE, GPIO1_OE register's software interface, set direction(in/out/bidirection) of GPIO

```
void smtGPIOSetDir (smtUint8 channel, smtUint8 gpioNum, smtUint32 mode);  
void smtGPIOGetDir (smtUint8 channel, smtUint8 gpioNum, smtUint32 *mode);
```

Parameters

channel

GPIO port number

gpioNum

GPIO port bit number

mode

In/output mode

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.3. smtGPIOSetData()/smtGPIOGetData()

Description

GPIO GPIO0_IN, GPIO0_OUT, GPIO1_IN, GPIO1_OUT register's software interface, write data or read data from GPIO controller

```
void smtGPIOSetData (smtUInt8 channel, smtUInt8 gpioNum, smtUInt32 data);  
void smtGPIOGetData (smtUInt8 channel, smtUInt8 gpioNum, smtUInt32 *data);
```

Parameters

channel

GPIO port number

gpioNum

GPIO port bit number

data

data

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

None

3.2.4. smtGPIOGetIntStatus ()

Description

GPIO port's gpio bit number interrupt status read

```
void smtGPIOGetIntStatus (smtUint8 channel, smtUint8 gpioNum, smtUint32 *gpioStatus);
```

Parameters

channel

GPIO port number

gpioNum

GPIO port bit number

gpioStatus

GPIO interrupt status of GPIO bit number

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
None
```

3.2.5. smtGPIOIntClear()

Description

GPIO0_INTSTAT, GPIO1_INTSTAT register's software interface, clear GPIO interrupt pending register

```
void smtGPIOIntClear (smtUInt8 channel, smtUInt8 gpioNum);
```

Parameters

channel

GPIO port number

gpioNum

GPIO port bit number

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
Refer to peri/gpio.c
```

3.2.6. smtGPIOIntEnable()

Description

GPIO0_INTEN, GPIO1_INTEN register's software interface, make enable/disable interrupt of GPIO

```
void smtGPIOIntEnable (smtUInt8 channel, smtUInt8 enable);
```

Parameters

channel

GPIO port

enable

GPIO port interrupt en/disable flag

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.7. smtGPIOSetIntProperty()/smtGPIOGetIntProperty()

Description

GPIO0_INTLEVEL, GPIO0_INTPOL, GPIO0_INTBEDGE, GPIO0_INTLEVEL, GPIO0_INTPOL, GPIO0_INTBEDGE, register's software interface. Set GPIO's interrupt level, interrupt polarity, interrupt trigger mode.

```
void smtGPIOSetIntProperty (smtUInt8 channel, GPIO_STRUCT gpio);  
void smtGPIOGetIntProperty (smtUInt8 channel, GPIO_STRUCT *gpio);
```

Parameters

channel

GPIO port

gpio

```
typedef struct  
{  
    smtUInt32 intStatus;           // GPIO interrupt status  
    smtUInt32 intNum;             // GPIO interrupt number  
    smtUInt32 intLevel;          // GPIO interrupt level  
    smtUInt32 intPolarity;        // GPIO interrupt polarity  
    smtUInt32 intBothEdge;        // GPIO interrupt both edge  
  
    smtUInt8  gpioNum;            // GPIO number  
} GPIO_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.3. TIMER

3.3.1. smtTIMERSetMode() / smtTIMERGetMode()

Description

Timer control register manipulate function

```
void smtTIMERSetMode (TIMER_STRUCT timer, TIMER_SEL timerSel);
void smtTIMERGetMode(TIMER_STRUCT *timer, TIMER_SEL timerSel);
```

Parameters

timer

```
typedef struct
{
    smtUInt32 data;           // Timer data register value
    smtUInt16 prescale;      // Timer prescaler value

    smtUInt8 opMode;         // Timer operation mode
    smtBoolean timerClear;   // Timer counter register clear
    smtBoolean timerEn;     // Timer enable
} TIMER_STRUCT;
```

timerSel

Timer channel

Enumerator	Description
TIMER_SELO	Timer channel 0
TIMER_SEL1	Timer channel 1
TIMER_SEL2	Timer channel 2
TIMER_SEL3	Timer channel 3

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

Refer to peri/timerpwm.c

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3.3.2. smtTIMERClrCount()

Description

Timer counter register clear function

```
void smtTIMERClrCount(smtUInt8 timerCh);
```

Parameters

timerCh

Timer channel

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

Refer to peri/timerpwm.c

3.3.3. smtTIMEROperation ()

Description

Timer operation(en/disable) function

```
void smtTIMEROperation (TIMER_STRUCT timer, TIMER_SEL timerSel);
```

Parameters

timer

```
typedef struct
{
    smtUInt32 data;           // Timer data register value
    smtUInt16 prescale;      // Timer prescaler value

    smtUInt8 opMode;         // Timer operation mode
    smtBoolean timerClear;   // Timer counter register clear
    smtBoolean timerEn;      // Timer enable
} TIMER_STRUCT;
```

timerSel

Timer channel

Enumerator	Description
TIMER_SEL0	Timer channel 0
TIMER_SEL1	Timer channel 1
TIMER_SEL2	Timer channel 2
TIMER_SEL3	Timer channel 3

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.4. VIC

3.4.1. EnableVIC()/DisableVIC()

Description

Write data or Read data form GPIO0/1

```
void EnableVIC(smtUInt32 Polarity, smtUInt32 Level, smtUInt32 IntMod);  
void DisableVIC(void);
```

Parameters

Polarity

Polarity of 32 interrupt source

Level

Interrupt event mode, level, edge mode

IntMod

Interrupt mode, IRQ/FIQ mode

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.2. RequestIRQ()/ReleaseIRQ()

Description

Write data or Read data form GPIO0/1

```
void RequestIRQ(smtUInt32 IRQ, ISRType ISR);  
void ReleaseIRQ(smtUInt32 IRQ);
```

Parameters

IRQ

IRQ number to Requested or Released

ISR

Interrupt handler to registry to VIC function table

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.3. RequestIRQ()/ReleaseIRQ()

Description

Write data or Read data form GPIO0/1

```
void RequestIRQ(smtUInt32 IRQ, ISRType ISR);  
void ReleaseIRQ(smtUInt32 IRQ);
```

Parameters

IRQ

IRQ number to Requested or Released

ISR

Interrupt handler to registry to VIC function table

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.4. EnableIRQ()/DisableIRQ()

Description

Enable/Disable IRQ of VIC

```
void EnableIRQ(smtUInt32 IRQ);  
void DisableIRQ(smtUInt32 IRQ);
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.5. AckIRQ()

Description

Clear interrupt source pending register for edge interrupt

```
void AckIRQ(smtUInt32 IRQ)
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.6. InterruptHandler()

Description

System IRQ top interrupt handler

```
void InterruptHandler(void);
```

Parameters

none.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.4.7. UndefHandler()

Description

System undefined instruction handler

```
void UndefHandler(int IRQ);
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.5. WDT

3.5.1. smtWDTSetMode () / smtWDTGetMode ()

Description

WDT control register manipulate function

```
void smtWDTSetMode(WDT_STRUCT wdt);  
void smtWDTGetMode(WDT_STRUCT *wdt);
```

Parameters

wdt

```
typedef struct  
{  
    smtUInt16 preScale;           // WDT prescaler register value  
    smtUInt16 reloadValue;       // WDT load register value  
  
    smtUInt8 div;                // WDT divide value  
    smtBoolean clkSel;           // WDT clock select  
    smtBoolean intEn;            // WDT interrupt en/disable  
    smtBoolean rstEn;            // WDT reset en/disable  
    smtBoolean wdtEn;            // WDT en/disable  
} WDT_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.5.2. smtWDTGetCount ()

Description

WDT counter register read function

```
void smtWDTGetCount(smtUint16 *count);
```

Parameters

count

WDT counter register value

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.5.3. smtWDTClearISR () / smtWDTGetISR ()

Description

WDT interrupt status register manipulate function

```
void smtWDTSetMode(void);  
void smtWDTGetMode(smtUInt8 *intrStatus);
```

Parameters

intrStatus

WDT interrupt status

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

Refer to peri/wdt.c

4. COMMUNICATION DEVICES

4.1. I2C

4.1.1. smtI2CSetCtrl() / smtI2CGetCtrl()

Description

I2C 통신을 위한 clk, operation mode, interrupt/ack enable 등의 정보 set/get하는 함수

```
void smtI2CSetCtrl (I2C_CTRL_STRUCT i2cCtrl);  
Void smtI2CGetCtrl (I2C_CTRL_STRUCT i2cCtrl);
```

Parameters

i2cCtrl

I2c 설정 구조체

```
typedef struct  
{  
    smtUInt16 txPrescale;           // I2C prescale value  
    smtUInt8 intPendFlag;          // I2C interrupt pending flag  
    smtUInt8 opMode;               // I2C operation mode (Tx/Rx)  
    smtUInt8 ackEn;                // I2C acknowledge enable  
    smtUInt8 txRxIntEn;            // I2C interrupt en/disable  
} I2C_CTRL_STRUCT;
```

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

None.

4.1.2. smtI2CGetStatus()

Description

이 함수는 I2C의 status 레지스터 정보를 가져온다.

```
void smtI2CGetStatus (I2C_STATUS_STRUCT i2cStat);
```

Parameters

i2cStat

I2c 설정 구조체

```
typedef struct
{
    smtBoolean outputEn;           // I2C output en/disable
    smtBoolean startStop;         // I2C start/stop
    smtBoolean busBusy;           // I2C bus busy/not busy
    smtBoolean arbitSta;          // I2C arbitration status
    smtBoolean ackValue;          // I2C receive ack status
    smtBoolean busError;          // I2C bus error status
} I2C_STATUS_STRUCT;
```

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

None.

4.1.3. smtI2CSWReset()

Description

이 함수는 I2C soft reset한다.

```
void smtI2CSWReset (void);
```

Parameters

none

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.4. smtI2CEnable()

Description

이 함수는 I2C의 동작을 en/disable한다.

```
void smtI2CEnable (smtUint8 enable);
```

Parameters

enable

I2C en/disable flag

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

None.

4.1.5. smtI2CInterruptWait()

Description

이 함수는 I2C의 pending flag가 발생하기를 기다린다.

```
void smtI2CInterruptWait (void);
```

Parameters

void.

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.6. smtI2CInterruptPendingClear()

Description

이 함수는 I2C의 pending flag를 clear 한다.

```
void smtI2CInterruptPendingClear (void);
```

Parameters

void

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.7. smtI2CInterruptPended()

Description

이 함수는 I2C pending flag를 return 한다.

```
smtInt32 smtI2CInterruptPended (void);
```

Parameters

void

Return Value

I2C pending status

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.8. smtI2CWrite()

Description

이 함수는 I2C를 통해 1byte 의 데이터 전송.

```
smtUint32smtI2CWrite(  
    smtUint8 addr,  
    smtUint8 subAddr,  
    smtUint8 data  
);
```

Parameters

addr

slave device address

subAddr

slave device subaddress

data

전송할 데이터

Return Value

SMT_SUCCESS/ SMT_ERROR

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

None.

4.1.9. smtI2CRead()

Description

이 함수는 I2C를 통해 slave device에서 데이터를 읽어들인다.

```
smtUInt32smtI2CRead (  
    smtUInt8 addr,  
    smtUInt8 subAddr,  
    smtUInt8 *data  
);
```

Parameters

addr

slave device address

subAddr

slave device subaddress

data

읽어들인 데이터

Return Value

SMT_SUCCESS/ SMT_ERROR

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

None.

4.1.10. smtI2CInit()

Description

이 함수는 I2C를 초기화한다.

```
void smtI2CInit (smtUint16 prescaler);
```

Parameters

prescaler

I2C prescale value

Return Value

void

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

None.

4.2. I2S

4.2.1. I2S_TRANS_DIR

Description

I2S transfer direction enumerated type

```
typedef enum
{
    I2S_DIR_RX,
    I2S_DIR_TX,
    I2S_DIR_32B    = 0xFFFF
} I2S_TRANS_DIR;
```

Parameters

I2S_TRANS_DIR

Type	Description
I2S_DIR_RX	I2S RX mode
I2S_DIR_TX	I2S TX mode

Return value

None

Remarks

None

Requirements

Header: i2s_pre_drv.h

Source: none

Library: none

4.2.2. I2S_DATA_WIDTH

Description

I2S data bits width enumerated type

```
typedef enum
{
    I2S_DW_08BIT,
    I2S_DW_16BIT,
    I2S_DW_24BIT,
    I2S_DW_32BIT,
    I2S_DW_32B      = 0xFFFF
} I2S_DATA_WIDTH;
```

Parameters

I2S_DATA_WIDTH

Type	Description
I2S_DW_08BIT	I2S 8bits mode
I2S_DW_16BIT	I2S 16bits mode
I2S_DW_24BIT	I2S 24bits mode
I2S_DW_32BIT	I2S 32bits mode

Return value

None

Remarks

None

Requirements

Header: i2s_pre_drv.h

Source: none

Library: none

4.2.3. I2S_FIFO_TH

Description

I2S FIFO threshold value enumerated type

```
typedef enum
{
    I2S_FIFO_TH00,
    I2S_FIFO_TH01,
    I2S_FIFO_TH02,
    I2S_FIFO_TH03,
    I2S_FIFO_TH04,
    I2S_FIFO_TH05,
    I2S_FIFO_TH06,
    I2S_FIFO_TH07,
    I2S_FIFO_TH08,
    I2S_FIFO_TH09,
    I2S_FIFO_TH10,
    I2S_FIFO_TH11,
    I2S_FIFO_TH12,
    I2S_FIFO_TH13,
    I2S_FIFO_TH14,
    I2S_FIFO_TH15,
    I2S_FIFO_32B = 0xFFFF
} I2S_FIFO_TH;
```

Parameters

I2S_FIFO_TH

Type	Description
I2S_FIFO_TH00~15	I2S FIFO threshold value 0 ~ 15

Return value

None

Remarks

None

Requirements

Header: i2s_pre_drv.h

Source: none

Library: none

4.2.4. I2S_CLOCK_MODE

Description

I2S Clock control register wrapper structure

```
typedef struct
{
    smtUint16      DTORatio;
    smtBoolean     ClkMaster;
    smtBoolean     LRClkInv;
    smtBoolean     MClkOn;
    smtBoolean     MClkOE;
} I2S_CLOCK_MODE;
```

Parameters

DTORatio

Internal 18bits Digital oscillator of I2S ratio

ClkMaster

I2S master enable bits

LRClkInv

I2S L/R clock invert enable bits

MClkOn

Internal 18bits Digital oscillator of I2S enable bits

MClkOE

Internal 18bits Digital oscillator of I2S clock out enable bits

Return value

None

Remarks

None

Requirements

Header: i2s_pre_drv.h

Source: none

Library: none

4.2.5. I2S_CONTROL

Description

I2S control register wrapper structure

```
typedef struct
{
    smtUInt8      WLen;
    smtUInt8      FifoTH;
    smtBoolean     Enable;
    smtBoolean     Reset;
    smtBoolean     DMAIntEn;
    smtBoolean     FifoOURIntEn;
    smtBoolean     Ljust;
} I2S_CONTROL;
```

Parameters

WLen

I2S bit width, refer to **I2S_DATA_WIDTH** enum

FifoTH

I2S fifo threshold value, refer to **I2S_FIFO_TH** enum

Enable

I2S RX/TX enable bit

Reset

I2S reset bit

DMAIntEn

Don't care, You don't need this bit

FifoOURIntEn

FIFO underrun/overflow interrupt enable bits

Ljust

Set is Left justified mode enable, Clear is I2S mode

Return value

None

Remarks

None

Requirements

Header: i2s_pre_drv.h

Source: none

Library: none

4.2.6. I2S_STATUS_DAT

Description

I2S status register wrapper structure

```
typedef struct
{
    smtBoolean      DMAInt;
    smtBoolean      FIFOURInt;
    smtUInt8        FIFODepth;
    smtBoolean      FIFOL;
} I2S_STATUS_DAT;
```

Parameters

DMAInt

Don't care, You don't need this bit

FIFOURInt

I2S overrun/underrun interrupt status bit

FIFODepth

I2S FIFO depth status bits

FIFOL

I2S FIFO level status bits

Return value

None

Remarks

None

Requirements

Header: i2s_pre_drv.h

Source: none

Library: none

4.2.7. smtSetClockMode()/smtGetClockMode()

Description

I2S_CLKCTRL 의 software interface

```
void smtSetClockMode(I2S_CLOCK_MODE *pclkMode)
void smtGetClockMode(I2S_CLOCK_MODE *pclkMode)
```

Parameters

pclkMode

I2S_CLKCTRL register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

Refer to test code (peri\comm\i2s.c)

4.2.8. smtI2SFIFOClear()

Description

I2S_STATUS RX/TX status clear bis's software interface

```
void smtI2SFIFOClear(I2S_TRANS_DIR dir)
```

Parameters

dir

I2S_DIR_RX is I2S RX FIFO clear, I2S_DIR_TX is I2S TX FIFO clear,

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

Refer to test code (peri\comm\i2s.c)

none.

4.2.9. smtI2SFIFOReset()

Description

I2S_CTRL의 RX/TX reset bit software interface

```
void smtI2SFIFOReset(I2S_TRANS_DIR dir)
```

Parameters

[dir](#)

I2S_DIR_RX is I2S RX FIFO reset, I2S_DIR_TX is I2S TX FIFO reset,

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

Refer to test code (peri\comm\i2s.c)

4.2.10. smtSetI2SMode()/smtGetI2SMode()

Description

I2S_CTRL's mode bits software interface

```
void smtSetI2SMode(I2S_TRANS_DIR dir, I2S_CONTROL *pcontrol)
```

Parameters

dir

I2S_DIR_RX is I2S RX FIFO reset, I2S_DIR_TX is I2S TX FIFO reset,

pcontrol

I2S_CTRL's DMAIntEn, FifoOURIntEn, WLen, Ljust, FifoTH 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

Refer to test code (peri\comm\i2s.c)

4.2.11. smtI2SEnable()

Description

I2S_CTRL's RX/TX enable bit software interface

```
void smtI2SEnable(I2S_TRANS_DIR dir, smtUint32 enable)
```

Parameters

dir

Direction of I2S interface, I2S_DIR_RX is RX mode, I2S_DIR_TXs TX mode.

enable

true is enable, false is disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

Refer to test code (peri\comm\i2s.c)

4.2.12. smtI2SGetStatus()

Description

I2S_STATUS's RX/TX status bit software interface

```
void smtI2SGetStatus(I2S_TRANS_DIR dir, I2S_STATUS_DAT *pstatus)
```

Parameters

dir

Direction of I2S interface, I2S_DIR_RX is RX mode, I2S_DIR_TXs TX mode.

pstatus

I2S_STATUS register 1:1 mapping layer per RX/TX

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

Refer to test code (peri\comm\i2s.c)

4.2.13. `smtI2SsetData()/smtI2SgetData()`

Description

I2S_STATUS's RX/TX status bit software interface

```
void smtI2SsetData(smtUint32 data)
smtUint32          smtI2SgetData(void)
```

Parameters

data

write data to I2S_DATA register

Return Value

Read data from I2S_DATA register.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: `i2s_pre_drv.h`

Source: `i2s_pre_drv.c`

Library: none

Example Code

Refer to test code (`peri\comm\i2s.c`)

4.3. SPI

4.3.1. smtSPISetMode() / smtSPIGetMode()

Description

이 함수는 SPI의 모드를 set/get 한다.

```
void smtSPISetMode (smtUInt8 channel, SPI_CON_STRUCT spiCtrl);  
void smtSPIGetMode (smtUInt8 channel, SPI_CON_STRUCT *spiCtrl);
```

Parameters

channel

SPI channel selection

spiCtrl

SPI control 구조체

```
typedef struct  
{  
    smtUInt8 spiEnable;           // SPI en/disable  
    smtUInt8 masterSel;          // SPI master/slave select  
    smtUInt8 polaritySel;        // SPI clock polarity setting  
    smtUInt8 phaseSel;           // SPI clock phase setting  
} SPI_CON_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

None.

4.3.2. smtSPISetPrescaler()

Description

이 함수는 SPI의 speed(prescale)를 설정한다.

```
void smtSPISetPrescaler (  
    smtUInt8 channel,  
    smtUInt32 preVal,  
    smtUInt32 divVal  
);
```

Parameters

channel

SPI channel selection

preVal

SPI prescaler value

divVal

SPI divider value

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

None.

4.3.3. smtSPIGetStatus()

Description

이 함수는 SPI의 상태 정보를 읽어온다.

```
void smtSPIGetStatus (smtUInt8 channel, SPI_STATUS_STRUCT *spiStatus);
```

Parameters

channel

spi channel selection

spiStatus

SPI 상태 정보 구조체

```
typedef struct
{
    smtUInt8 txFifoLevel;           // SPI Tx FIFO level
    smtUInt8 rxFifoLevel;           // SPI Rx FIFO level
    smtUInt8 txFifoFull;            // SPI Tx FIFO full flag
    smtUInt8 txFifoEmpty;           // SPI Tx FIFO empty flag
    smtUInt8 rxFifoFull;            // SPI Rx FIFO full flag
    smtUInt8 rxFifoEmpty;           // SPI Rx FIFO empty flag
    smtUInt8 busy;                  // SPI bus transfer status
} SPI_STATUS_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

None.

4.3.4. smtSPISetIntDMA() / smtSPIGetIntDMA()

Description

이 함수는 SPI의 interrupt en/disable 및 DMA en/disable 등을 set/get 한다.

```
void smtSPISetIntDMA (smtUInt8 channel, SPI_INTDMA_STRUCT spiIntDMA);
void smtSPIGetIntDMA (smtUInt8 channel, SPI_INTDMA_STRUCT *spiIntDMA);
```

Parameters

channel

SPI channel selection

spiIntDMA

SPI interrupt, DMA 설정 구조체

```
typedef struct
{
    smtUInt8 txFifoIntEn;           // Tx FIFO interrupt enable
    smtUInt8 rxFifoIntEn;           // Rx FIFO interrupt enable
    smtUInt8 rxTimeOutIntEn;        // Rx timeout interrupt enable
    smtUInt8 rxOverRunIntEn;        // Rx overrun interrupt enable
    smtUInt8 txDMAEn;               // Tx DMA request enable
    smtUInt8 txDMAlevel;            // Tx DMA request level
    smtUInt8 rxDMAEn;               // Rx DMA request enable
    smtUInt8 rxDMAlevel;            // Rx DMA request level
} SPI_INTDMA_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

None.

4.3.5. smtSPIGetIntStatus()

Description

이 함수는 SPI의 인터럽트 정보를 읽어온다.

```
void smtSPIGetIntStatus (smtUInt8 channel, SPI_INTSTA_STRUCT *spiIntDMA);
```

Parameters

channel

SPI channel selection

spiIntDMA

SPI 인터럽트 상태 구조체

```
typedef struct
{
    smtUInt8 rToutIntClr;           // Rx timeout interrupt clear
    smtUInt8 rOverIntClr;          // Rx overrun interrupt clear
    smtUInt8 txFifoIntSta;          // Tx FIFO interrupt status
    smtUInt8 rxFifoIntSta;          // Rx FIFO interrupt status
    smtUInt8 rxTimeOutIntSta;       // Rx timeout interrupt status
    smtUInt8 rxOverRunIntSta;       // Rx overrun interrupt status
} SPI_INTSTA_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

None.

4.3.6. smtSPIClrIntStatus()

Description

이 함수는 SPI의 인터럽트 상태를 clear 한다.

```
void smtSPIClrIntStatus (smtUInt8 channel, SPI_INTSTA_STRUCT spitIntDMA);
```

Parameters

channel

SPI channel selection

spiIntDMA

SPI 인터럽트 상태 구조체

```
typedef struct
{
    smtUInt8 rToutIntClr;           // Rx timeout interrupt clear
    smtUInt8 rOverIntClr;          // Rx overrun interrupt clear
    smtUInt8 txFifoIntSta;          // Tx FIFO interrupt status
    smtUInt8 rxFifoIntSta;          // Rx FIFO interrupt status
    smtUInt8 rxTimeOutIntSta;       // Rx timeout interrupt status
    smtUInt8 rxOverRunIntSta;       // Rx overrun interrupt status
} SPI_INTSTA_STRUCT;
```

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

None.

4.3.7. smtSPIONebyteWrite()

Description

이 함수는 SPI를 통해 1byte 전송한다.

```
void smtSPIONebyteWrite (smtInt8 channel, smtInt8 data);
```

Parameters

channel

SPI channel selection

data

전송할 1 바이트 데이터

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.8. smtSPIOnebyteRead()

Description

이 함수는 SPI를 통해 1byte를 읽어온다.

```
smtUInt8 smtSPIOnebyteRead (smtInt8 channel);
```

Parameters

channel

SPI channel selection

Return Value

smtUInt8

읽어온 1 바이트 데이터

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

None.

4.3.9. smtSPIInit()

Description

이 함수는 SPI를 통해 1byte 전송한다.

```
void smtSPIInit (  
    smtInt8 channel,  
    smtUInt32 PreVal,  
    smtUInt32 DivVal  
);
```

Parameters

channel

SPI channel selection

PreVal

SPI prescaler value.

DivVal

SPI divider value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

None.

4.4. UART

4.4.1. UartChannel

Description

uart channel enumeration value.

```
typedef enum
{
    UART_CHANNEL0,
    UART_CHANNEL1,
    UART_CHANNEL2,
    UART_CHANNEL3,
    MAX_UART_CHANNEL
} UartChannel;
```

Parameters

none.

Return value

none.

Remarks

none.

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

```
None
```

4.4.2. UartBaudRate

Description

uart baudrate enumeration value.

```
typedef enum
{
    BAUD_4800          = 4800,
    BAUD_9600          = 9600,
    BAUD_14400         = 14400,
    BAUD_19200         = 19200,
    BUAD_31257         = 31257,
    BAUD_38400         = 38400,
    BAUD_115200        = 115200
} UartBaudRate;
```

Parameters

none.

Return value

None

Remarks

none

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

```
None
```

4.4.3. UartMode

Description

uart pre driver 함수(smtUartSetMode/ smtUartGetMode) 호출할 때 register 및 operation mode를 나타내는 enumeration value.

```
typedef enum
{
    GET_UARTMASTER,
    SET_UARTMASTER,
    GET_UARTSTATUS,
    GET_UARTBRD,
    SET_UARTBRD,
    GET_UARTRXTIMEOUT,
    SET_UARTRXTIMEOUT
} UartMode;
```

Parameters

GET_UARTMASTER/ SET_UARTMASTER

uart master register read/write.

GET_UARTSTATUS/ SET_UARTSTATUS

uart status register read/write.

GET_UARTBRD/ SET_UARTBRD

uart Baudrate register read/write.

GET_UARTRXTIMEOUT/ SET_UARTRXTIMEOUT

uart rx timeout register read/write.

Return value

None

Remarks

none

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

None

4.4.4. UartParity

Description

uart parity bit enumeration value.

```
typedef enum
{
    PARITY_DISABLE,
    PARITY_EVEN = 2,
    PARITY_ODD = 3
} UartParity;
```

Parameters

PARITY_DISABLE

disable parity bit.

PARITY_EVEN

even parity.

PARITY_ODD

odd parity.

Return value

None

Remarks

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

None

4.4.5. UartStopBit

Description

uart stop bit enumeration value.

```
typedef enum
{
    ONE_STOP_BIT,
    TWO_STOP_BIT
} UartStopBit;
```

Parameters

ONE_STOP_BIT

one stop bit.

TWO_STOP_BIT

two stop bit.

Return value

None

Remarks

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

None

4.4.6. UartDataBit

Description

uart data bit enumeration value.

```
typedef enum
{
    DATA_7BIT,
    DATA_8BIT
} UartDataBit;
```

Parameters

DATA_7BIT

uart data 7bit.

DATA_8BIT

uart data 8bit.

Return value

None

Remarks

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

None

4.4.7. UartMaster

Description

uart master register wrapper structure.

```
typedef struct
{
    smtBoolean    uartEn;
    smtBoolean    intEn;
    smtBoolean    rxTimeoutEn;
    smtBoolean    swReset;
    smtBoolean    dmaReqEn;
    smtUInt8 parity;
    smtUInt8 dataBit;
    smtUInt8 stopBit;
    smtBoolean    loopbackEn;
    smtBoolean    txIntEn;
    smtBoolean    rxIntEn;
    smtUInt8    txWaterLev;
    smtUInt8    rxWaterLev;
}UartMaster;
```

Parameters

uartEn

uart enable bit.

intEn

uart interrupt enable.

rxTimeoutEn

uart receive timeout interrupt enable.

swReset

uart sw reset.

dmaReqEn

uart dma request enable bit.

parity

uart parity bit selection bit.

dataBit

uart data bit selection bit.

stopBit

uart stop selection bit.

loopbackEn

uart loopback enable bit.

txIntEn

uart tx interrupt enable bit.

rxIntEn

uart rx interrupt enable bit.

txWaterLev

uart tx interrupt water level.

rxWaterLev

uart rx interrupt water level.

Return value

None

Remarks

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

```
None
```

4.4.8. UartStatus

Description

uart status register wrapper structure.

```
typedef struct
{
    smtBoolean    interrupt;
    smtUInt8      maxFifoDepth;
    smtBoolean    txBusy;
    smtBoolean    rxBusy;
    smtBoolean    parityErr;
    smtBoolean    frameErr;
    smtBoolean    overrunErr;
    smtBoolean    rxTimeout;
    smtBoolean    txFifoInt;
    smtBoolean    rxFifoInt;
    smtBoolean    txFifoDMAReq;
    smtBoolean    rxFifoDMAReq;
    smtUInt8      txFifoCnt;
    smtBoolean    rxFifoDMAReq;
    smtUInt8      rxFifoCnt;
}UartStatus;
```

Parameters

interrupt

uart interrupt status.

maxFifoDepth

uart max fifo depth.

txBusy

uart tx busy status.

rxBusy

uart rx busy status.

parityErr

uart parity error status.

frameErr

uart frame error status.

overrunErr

uart overrun error status.

rxTimeout

uart rx timeout status.

txFifoInt

uart tx fifo interrupt status.

rxFifoInt

uart rx fifo interrupt status.

txFifoDMAReq

uart tx fifo DMA request status.

rxFifoDMAReq

uart rx fifo DMA request status.

txFifoCnt

uart tx fifo counter.

rxFifoCnt

uart rx fifo counter.

Return value

None

Remarks

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

```
None
```

4.4.9. UartStruct

Description

uart top structure including all basic uart struct. This structure is used to call uart pre-driver functions like smtUartGetMode/smtUartSetMode.

```
typedef struct
{
    UartMaster      uartMaster;
    UartStatus      uartStatus;
    smtUInt32       uartRxTimeout;
    smtUInt16       uartBaudGen;
} UartStruct;
```

Parameters

uartMaster

wrapper structure of uart master register.

uartStatus

wrapper structure of uart status register.

uartRxTimeout

This member contains uart rx timeout counter register value.

uartBaudGen

This member contains uart baudrate register value.

Return value

None

Remarks

Requirements

Header: uart_pre_drv.h

Source: none

Library: none

Example Code

None

4.4.10. UartConfig

Description

UART 초기화.

```
typedef struct
{
    smtUInt8 baudRate;
    smtUInt8 parity;
```

```
smtUInt8 stopBit;
smtUInt8 dataBit;
smtUInt8 rxWaterLev;
smtUInt8 txWaterLev;
smtBoolean rxTimeOutEn;
smtUInt16 rxTimeOut;
UartChannel uartCh;
}UartConfig;
```

Parameters

baudRate

UART baudrate 설정.

parity

UART parity bit 설정.

stopBit

UART stop bit 설정.

dataBit

UART data bit width 설정.

rxWaterLev

UART receive FIFO water level 설정.

txWaterLev

UART transmit FIFO water level 설정.

rxTimeoutEn

UART receive time out enable 설정.

rxTimeOut

UART receive time out count 설정.

uartCh

UART의 채널을 설정.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires Skywalker firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

example code here

4.4.11. smt2UartInit()

Description

UART 초기화.

```
smtBoolean  
smt2UartInit(  
    UartConfig uartCfg  
);
```

Parameters

uartCfg

UART 초기화 정보를 담고 있는 구조체.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

none.

4.4.12. smt2UartDeInit ()

Description

initialize 한 UART 채널을 disable.

```
void  
smt2UartDeInit(  
    void  
)
```

Parameters

none.

Return Value

none

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none
```

4.4.13. smt2UartPutStr()

Description

UART를 통해 문자열을 출력.

```
smtBoolean  
smt2UartPutStr(  
    smtInt8 *pData  
);
```

Parameters

pData

uart를 통해 출력할 문자열.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

입력 문자열의 마지막은 '\0' 여야 한다.

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.14. smt2UartGetStr()

Description

UART를 통해 문자열을 입력 받는다.

```
smtBoolean  
smt2UartGetStr(  
    smtUInt8 *pData,  
    smtUInt8 dataCnt  
);
```

Parameters

pData

uart를 통해 출력할 문자열.

dataCnt

읽어들일 문자열의 개수.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.15. smt2UartPrint()

Description

UART를 통해 포맷 형식의 문자열을 출력한다.

```
smtBoolean  
smt2UartPrint(  
    int dispLvl,  
    char *fmt,  
    [argument]...  
);
```

Parameters

dispLvl

출력 레벨 설정값.

fmt

출력 포맷 설정.

[argument]...

선택적 인자들.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.16. smt2UartPutCh()

Description

UART를 통해 한 개의 문자를 출력한다.

```
smtBoolean  
smt2UartPutCh(  
    smtInt8 data  
);
```

Parameters

data

출력할 문자 데이터.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.17. smt2UartGetCh()

Description

UART를 통해 한 개의 문자를 입력 받는다.

```
smtUInt8  
smt2UartGetCh(  
    smtUInt32 waitTime  
);
```

Parameters

waitTime

time out value.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.18. smtUartSetMode()

Description

UART master/status/baud generation/RX timeout register 변경.

```
smtBoolean  
smtUartSetMode(  
    UartChannel uartCh,  
    UartStruct *pUart,  
    UartMode mode  
);
```

Parameters

uartCh

UART channel 선택.

pUart

register setting 정보를 담고 있는 구조체 포인터.

mode

setting 할 target register를 선택.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.19. smtUartGetMode()

Description

UART master/status/baud generation/RX timeout register 값을 읽어서 인수로 넘어온 구조체에 저장한다..

```
smtBoolean  
smtUartGetMode(  
    UartChannel uartCh,  
    UartStruct *pUart,  
    UartMode mode  
);
```

Parameters

uartCh

UART channel 선택.

pUart

register setting 정보를 저장할 구조체 포인터.

mode

읽어들일 target register를 선택.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.20. smtUartReadRxFifo()

Description

UART RX FIFO에서 1byte size의 데이터를 읽어 온다.

```
smtUInt8  
smtUartReadRxFifo(  
    UartChannel uartCh  
);
```

Parameters

uartCh

UART channel 선택.

Return Value

RX FIFO에 읽어 들인 1byte size의 데이터.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.21. smtUartWriteTxFifo()

Description

UART RX FIFO에서 1byte size의 데이터를 읽어 온다.

```
void  
smtUartWriteTxFifo(  
    UartChannel uartCh,  
    smtUInt8 txData  
);
```

Parameters

uartCh

UART channel 선택.

txData

TX FIFO에 쓸 데이터.

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.22. smtUartDataAvailable()

Description

UART RX FIFO의 count 값을 체크 하여 읽어 들일 데이터가 있는지를 검사한다.

```
smtUint32 smtUartDataAvailable(void);
```

Parameters

none.

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.23. uart multi channel post driver functions

Description

these functions support multi channel uart operation. all functions do the same job as older ones.

```
smtUint32 smt2UartMultiInit(UartConfig uartInitData);
smtUint32 smt2UartMultiDeInit(smtUint8 uartCh);
smtUint32 smt2UartMultiPutStr(smtUint8 uartCh, smtInt8 * str);
smtUint32 smt2UartMultiGetStr(smtUint8 uartCh, smtUint8 *pData, smtUint8 dataCnt);
smtUint32 smt2UartMultiPrint(smtUint8 uartCh, smtInt32 dispLvl, smtInt8
*format, ...);
smtUint32 smt2UartMultiPutCh(smtUint8 uartCh, smtUint8 ch);
smtUint8 smt2UartMultiGetCh(smtUint8 uartCh, smtUint32 waitTime);
smtUint8 smt2UartMultiDataValid(smtUint8 uartCh);
```

Parameters

none.

Return Value

none.

Remarks

smt2UartMultiDataValid() checks if uart rx fifo has a data to be recieved.

Requirement

Platform: Requires Skywalker firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
example code here
```

Confidential

4.5. USB

Not yet

Confidential

5. DISPLAY DEVICES

5.1. DM

5.1.1. DMPlane

Description

C/G/V plane enumeration value.

Parameters

none

Return Value

none.

Remarks

Type define	description
BACKGROUND_PLANE	Back ground plane.
CURSOR_PLANE	Cursor(Graphic0) plane
GRAPHIC_PLANE	Graphic1 plane.
VIDEO_PLANE	Video plane

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

```
example code here
```

5.1.2. DMGammaIdx

Description

Graphic/Video plane의 gamma table index를 위한 enumeration value.

Parameters

none

Return Value

Type define	description
GAMMA_IDXxx	gamma table index에 사용된다. xx(0~16).

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

example code here

5.1.3. DMCtrl

Description

DM(Display Module)의 LCDMCON register wrapper 구조체.

```
typedef struct
{
    smtBoolean prioritySel;    // Plane position
    smtBoolean errIntEn;      // error interrupt enable
    smtBoolean startIntEn;    // start interrupt enable
    smtBoolean endIntEn;      // end Interrupt enable
    smtBoolean swReset;       // SW reset
} DMCtrl;
```

Parameters

prioritySel

cursor/graphic/video plane의 우선순위 설정.

errIntEn

DM의 error interrupt 활성화 여부 설정.

startIntEn

start frame interrupt 활성화 여부 설정.

endIntEn

end frame interrupt 활성화 여부 설정.

swReset

DM을 리셋한다.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

```
example code here
```

Confidential

5.1.4. DMStatus

Description

DM(Display Module)의 LCDMSTS register wrapper 구조체.

```
typedef struct
{
    smtBoolean    evenFieldInt;    // Even field Interrupt
    smtBoolean    cDmaFifoErr;    // Cursor plane DMA fifo error
    smtBoolean    vDmaFifoErr;    // Video plane DMA fifo error
    smtBoolean    gDmaFifoErr;    // Graphic plane DMA fifo error
    smtBoolean    cFifoErr;       // Cursor plane fifo error
    smtBoolean    vFifoErr;       // Video plane fifo error
    smtBoolean    gFifoErr;       // Graphic plane fifo error
    smtBoolean    mixFifoErr;     // Mixer fifo error
    smtBoolean    endFrame;       // End of frame status
    smtBoolean    evenField;      // Interrupt status
} DMStatus;
```

Parameters

evenFieldInt

indicate even/odd field

cDmaFifoErr

cursor plane DMA FIFO OverRun.

vDmaFifoErr

video plane DMA FIFO OverRun.

gDmaFifoErr

graphic plane DMA FIFO OverRun.

cFifoErr

cursor plane FIFO UnderRun.

vFifoErr

video plane FIFO UnderRun.

gFifoErr

graphic plane FIFO UnderRun.

mixFifoErr

mixer FIFO UnderRun.

endFrame

start of frame

evenField

even field indication.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

```
example code here
```

Confidential

5.1.5. DMPlaneCtrl

Description

DM(Display Module)의 각 plane(cursor/graphic/video)의 control register wrapper 구조체.

```
typedef struct
{
    smtUInt16    sWidth;        // Width
    smtUInt16    sHeight;       // Height
    smtUInt8     pixFmt;        // Pixel format
    smtBoolean    xEn;          // C/V/Graphic plane enable
    smtBoolean    gammaEn;      // Gamma enable. Only for graphic/video plane.
    smtBoolean    vPlaneIFEn;   // video plane only.
    smtBoolean    vPlaneN2PUpEn; // video plane only.
} DMPlaneCtrl;
```

Parameters

sWidth

plane surface width.

sHeight

plane surface height.

pixFmt

plane의 픽셀 포맷 설정.

xEn

plane enable 설정.

gammaEn

plane gamma enable 설정.

vPlaneIFEn

interface video enable 설정.

vPlaneN2PUpEn

add 1 line each 5 line enable 설정(720x480 to 720x576)

Return Value

none.

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

example code here

Confidential

5.1.6. DMPlaneBlnd

Description

DM(Display Module)의 각 plane(cursor/graphic/video)의 blending register wrapper 구조체.

```
typedef struct
{
    smtUint32    colKey;        // Color key value
    smtUint8     alpha;        // Alpha-blending value
} DMPlaneBlnd;
```

Parameters

colKey

color key 설정.

alpha

global alpha blending 값 설정(0~255).

Return Value

none.

Remarks

color key 값은 기본적으로 RGB888 형태로 입력 해야 한다.

- ✓ source image format이 RGB565 인경우 흰색을 chromakey 하고자 한다면,
color key 값으로 0xFCF8FC 를 입력한다.
- ✓ plane에 gamma correction 이 이루어진 경우, gamma correction에 의해 변경된
RGB값으로 color key를 설정한다.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

```
example code here
```

5.1.7. DMPlaneBlndMode

Description

DM(Display Module)의 각 plane(cursor/graphic/video)의 blend mode register wrapper 구조체.

```
typedef struct
{
    smtUInt16      xPos;           //plane x position
    smtUInt16      yPos;           //plane y position
    smtBoolean      colKeyEn;       // Color key enable
    smtUInt8        blendMod;       // Blending mode select
} DMPlaneBlndMode;
```

Parameters

xPos

plane의 x position start 설정.(tv/lcd 상의 좌표)

yPos

plane의 y position start 설정. (tv/lcd 상의 좌표)

colKeyEn

color key enable 설정.

blendMod

alpha blending mode 설정.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

example code here

5.1.8. DMPlanePalette

Description

DM(Display Module)의 각 plane(cursor/graphic)의 palette memory register wrapper 구조체.

```
typedef struct
{
    smtUInt32    palData;        // Palette data
    smtUInt8     palAddr;        // Palette addr
} DMPlanePalette;
```

Parameters

palData

plane의 palette data 설정.

palAddr

plane의 palette index 설정.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

```
example code here
```

5.1.9. DMLcdCtrl

Description

DM(Display Module)의 Lcd control register wrapper 구조체.

```
typedef struct
{
    smtBoolean    lcdEn;           // LCD sync enable
    smtBoolean    lcdPwrEn;       // LCD/Video data out enable
    smtUInt8      lcdBpp;         // LCD pixel format
    smtBoolean    rbSwap;         // R/B swap ENABLE
    smtBoolean    invertWrEn;     // Invert write enable
    smtBoolean    invertVSync;    // Invert Vsync
    smtBoolean    invertHSync;    // Invert Hsync
    smtBoolean    invertPixClk;   // Invert pixel clock
    smtUInt8      pDiv;           // Phase divide
} DMLcdCtrl;
```

Parameters

lcdEn

Lcd sync enable 설정.

lcdPwrEn

Lcd/video data out enable 설정.

lcdBpp

Lcd bit per pixel 설정.

lcdSwap

R/B swap enable 설정.

invertWrEn

invert Write enable 설정.

invertVSync

invert VSync enable 설정.

invertHSync

invert VSync enable 설정.

invertPixClk

invert Pixel clock enable 설정.

pDiv

system clock divide 설정.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

```
example code here
```

Confidential

5.1.10. DMHSync0Ctrl/ DMHSync1Ctrl

Description

DM(Display Module)의 Lcd horizontal sync0/1 register wrapper 구조체.

```
typedef struct
{
    smtUInt8      tHBackPorch;    // Horizontal back porch
    smtUInt8      tHFrontPorch;   // Horizontal front porch
} DMHSync0Ctrl;

typedef struct
{
    smtUInt16     tHPulsWidth;     // Horizontal sync pulse width
    smtUInt16     tHDspPixPerLine; // display pixel per line
} DMHSync1Ctrl;
```

Parameters

tHBackPorch

Lcd HSync back porch 설정.

tHFrontPorch

Lcd HSync front porch 설정.

tHPulsWidth

Horizontal Sync Pulse Width 설정.

tHDspPixPerLine

Display Pixel per Line 설정.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

example code here

5.1.11. DMVSync0Ctrl/ DMVSync1Ctrl

Description

DM(Display Module)의 Lcd vertical sync0/1 register wrapper 구조체.

```
typedef struct
{
    smtUInt8      tVBackPorch;    // Vertical back porch
    smtUInt8      tVFrontPorch;   // Vertical front porch
} DMVSync0Ctrl;

typedef struct
{
    smtUInt16     tVDspPeriod;    // Line display period
    smtUInt8      tVPulsWidth;    // Vertical sync pulse width
} DMVSync1Ctrl;
```

Parameters

tVBackPorch

Lcd VSync back porch 설정.

tVFrontPorch

Lcd VSync front porch 설정.

tVPulsWidth

vertical Sync Pulse Width 설정.

tVDspPixPerLine

line display period 설정.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires Skywalker firmware platform

Header: lcd_pre_drv.h

Source: none.

Library: none.

Example Code

example code here

5.1.12. smtDMSetMasterCtrl()

Description

DM(Display Module)의 DMCON register를 설정한다.

```
void  
smtDMSetMasterCtrl(  
    DMCtrl maCtrl  
);
```

Parameters

maCtrl

DMCON register wrapper 구조체.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMCtrl dmCtrl;  
  
//1. set DM control  
dmCtrl.prioritySel    = 0x1;// priority select  
dmCtrl.swReset        = 0x0;// DM software reset  
dmCtrl.errIntEn       = 0x0;// enable DM error interrupt  
dmCtrl.intEn          = 0x1;// enable frame start interrupt  
smtDMSetMasterCtrl(dmCtrl);
```

5.1.13. smtDMGetMasterCtrl()

Description

DM(Display Module)의 DMCON register를 읽어 들인다.

```
void  
smtDMGetMasterCtrl(  
    DMCtrl *pMaCtrl  
);
```

Parameters

pMaCtrl

DMCON register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMCtrl dmMasterCtrl;  
smtDMGetMasterCtrl(&dmMasterCtrl);  
dmMasterCtrl.swReset = 0x1;  
smtDMSetMasterCtrl(dmMasterCtrl);
```

5.1.14. smtDMGetMasterStatus()

Description

DM(Display Module)의 LCDMSTS register를 읽어 들인다.

```
void  
smtDMGetMasterStatus(  
    DMStatus *pMaStat  
);
```

Parameters

pMaStat

LCDMSTS register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMStatus dmSts;  
smtDMGetMasterStatus(&dmSts); //read DM status register  
  
if(dmSts.startFrame == 1)  
{  
    if(dmSts.evenFieldInt == 1)  
        bChgOddFrame = SMT_TRUE;  
    else  
        bChgEvenFrame = SMT_TRUE;  
}
```

5.1.15. smtDMSetCCtrl()/smtDMGetCCtrl()

Description

DM(Display Module)의 cursor plane control register를 read/write.

```
void smtDMSetCCtrl(DMPlaneCtrl Ctrl);  
void smtDMGetCCtrl(DMPlaneCtrl *pCtrl);
```

Parameters

ctrl

cursor plane control register wrapper 구조체.

pCtrl

cursor plane control register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
//4. set plane control register  
cPlaneCfg.xEn      = 0x1;  
cPlaneCfg.pixFmt    = 0x3;//0:8BPP,1:RGB332 2:ARGB(1555) 3:RGB565  
cPlaneCfg.gammaEn   = 0x0;  
cPlaneCfg.sWidth    = CURSOR_WIDTH;  
cPlaneCfg.sHeight   = CURSOR_HEIGHT;  
smtDMSetCCtrl(cPlaneCfg);
```

5.1.16. smtDMSetCBlndMode()/smtDMGetCBlndMode()

Description

DM(Display Module)의 cursor plane blend mode register를 read/write.

```
void smtDMSetCBlndMode(DMPlaneBlndMode blndMode);
void smtDMGetCBlndMode(DMPlaneBlndMode *pBlndMode);
```

Parameters

blndMode

cursor plane blend mode register wrapper 구조체.

pBlndMode

cursor plane blend mode register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
cPlaneBlndMode.xPos = 0;
cPlaneBlndMode.yPos = 0;
cPlaneBlndMode.colKeyEn = 0x0; //dis/enable color key
cPlaneBlndMode.blendMod = 0x0; //0:no alpha , 2: global alpha, 3: pixel alpha
smtDMSetCBlndMode(cPlaneBlndMode);
```

5.1.17. smtDMSetCBlnd()/smtDMGetCBlnd()

Description

DM(Display Module)의 cursor plane blend register를 read/write.

```
void smtDMSetCBlnd(DMPlaneBlnd blnd);  
void smtDMGetCBlnd(DMPlaneBlnd *pblnd);
```

Parameters

blnd

cursor plane blend register wrapper 구조체.

pBlnd

cursor plane blend register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
//1. set chromakey/alpha value(BLEND)  
cPlaneBlnd.alpha = 0xFF;// global alpha vale  
cPlaneBlnd.colKey = 0xFFFFFFFF;  
smtDMSetCBlnd(cPlaneBlnd);
```

5.1.18. smtDMSetCBase()/smtDMGetCBase()

Description

DM(Display Module)의 cursor plane base address register를 read/write.

```
void smtDMSetCBase(smtUint16 planeXRef);  
smtUint16 smtDMGetCBase(void);
```

Parameters

planeXRef

cursor plane XRef value.

Return Value

none.

Remarks

plane source image의 width 와 plane의 width가 같은 경우 xRef는 0 이 된다.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtDMSetCBase(0x0);
```

5.1.19. smtDMSetCAddr()/smtDMGetCAddr()

Description

DM(Display Module)의 cursor plane start address register를 read/write.

```
void smtDMSetGAddr(smtUInt32 addr);  
smtUInt32 smtDMGetGAddr(void);
```

Parameters

addr

cursor plane start addr value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtUInt16 *frame0 = (smtUInt16 *)CURSOR_BASEADDR;  
smtDMSetCAddr((smtUInt32)frame0);
```

5.1.20. smtDMSetCPalette()/smtDMGetCPalette()

Description

DM(Display Module)의 cursor plane palette register를 read/write.

```
void smtDMSetCPalette(DMPlanePalette palette);  
void smtDMGetCPalette(DMPlanePalette *pPalette);
```

Parameters

palette

cursor plane palette register 정보를 담고 있는 구조체.

pPalette

cursor plane palette register 정보를 담고 있는 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
none.
```

5.1.21. smtDMSetGCtrl()/smtDMGetGCtrl()

Description

DM(Display Module)의 graphic plane control register를 read/write.

```
void smtDMSetGCtrl(DMPlaneCtrl ctrl);
void smtDMGetGCtrl(DMPlaneCtrl *pCtrl);
```

Parameters

ctrl

graphic plane control register wrapper 구조체.

pCtrl

graphic plane control register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMPlaneCtrl gPlaneCfg;

gPlaneCfg.xEn          = 0x1;
gPlaneCfg.pixFmt        = 0x4; //0:8BPP, 2:RGB565, 3:ARGB1555, 4:RGB888, 5:ARGB8888
gPlaneCfg.gammaEn       = 0x0;
gPlaneCfg.sWidth        = GRAPHIC_WIDTH;
gPlaneCfg.sHeight       = GRAPHIC_HEIGHT;

smtDMSetGCtrl(gPlaneCfg);
```

5.1.22. smtDMSetGBIndMode()/smtDMGetGBIndMode()

Description

DM(Display Module)의 graphic plane blend mode register를 read/write.

```
void smtDMSetGBIndMode(DMPlaneBlndMode blndMode);
void smtDMGetGBIndMode(DMPlaneBlndMode *pBlndMode);
```

Parameters

blndMode

graphic plane blend mode register wrapper 구조체.

pBlndMode

graphic plane blend mode register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMPlaneBlndMode gPlaneBlndMode;
gPlaneBlndMode.xPos = 0;
gPlaneBlndMode.yPos = 0;
gPlaneBlndMode.colKeyEn = 0x0;
gPlaneBlndMode.blendMod = 0x2; //0:no alpha , 2: global alpha, 3: pixel alpha
smtDMSetGBIndMode(gPlaneBlndMode);
```

5.1.23. smtDMSetGBInd()/smtDMGetGBInd()

Description

DM(Display Module)의 graphic plane blend register를 read/write.

```
void smtDMSetCBInd(DMPlaneBlnd blnd);
void smtDMGetCBInd(DMPlaneBlnd *pblnd);
```

Parameters

blnd

graphic plane blend register wrapper 구조체.

pBlnd

graphic plane blend register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
gPlaneBlnd.alpha = 0xFF;
gPlaneBlnd.colKey = 0xFFFFFFFF;
smtDMSetGBInd(gPlaneBlnd);
```

5.1.24. smtDMSetGBase()/smtDMGetGBase()

Description

DM(Display Module)의 graphic plane base address register를 read/write.

```
void smtDMSetCBase(smtUint16 planeXRef);  
smtUint16 smtDMGetCBase(void);
```

Parameters

planeXRef

graphic plane XRef value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtDMSetGBase(0x0);
```

5.1.25. smtDMSetGAddr()/smtDMGetGAddr()

Description

DM(Display Module)의 graphic plane start address register를 read/write.

```
void smtDMSetCAddr(smtUint32 addr);  
smtUint32 smtDMGetCAddr(void);
```

Parameters

addr

graphic plane start addr value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtUint32 *frame0 = (smtUint32 *)GRAPHIC_BASEADDR;  
smtDMSetGAddr((smtUint32)frame0);
```

5.1.26. smtDMSetGPalette()/smtDMGetGPalette()

Description

DM(Display Module)의 graphic plane palette register를 read/write.

```
void smtDMSetCPalette(DMPlanePalette palette);  
void smtDMGetCPalette(DMPlanePalette *pPalette);
```

Parameters

palette

graphic plane palette register wrapper 구조체.

pPalette

graphic plane palette register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
none.
```

5.1.27. smtDMSetGGamma()/smtDMGetGGamma()

Description

DM(Display Module)의 graphic plane gamma register를 read/write.

```
void smtDMSetGGamma(smtUInt32 addr, smtUInt32 gammaVal);  
smtUInt32 smtDMGetGGamma(smtUInt32 addr);
```

Parameters

addr

Gamma look up table index.

gammaVal

Gamma look up table index에 해당하는 gamma value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
none.
```

5.1.28. smtDMSetVCtrl()/smtDMGetVCtrl()

Description

DM(Display Module)의 video plane control register를 read/write.

```
void smtDMSetVCtrl(DMPlaneCtrl ctrl);
void smtDMGetVCtrl(DMPlaneCtrl *pCtrl);
```

Parameters

ctrl

video plane control register wrapper 구조체.

pCtrl

video plane control register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
vPlaneCfg.xEn      = 0x1;
vPlaneCfg.pixFmt    = 0x3;
vPlaneCfg.vPlaneIFEn = 0x0; // interlaced video source
vPlaneCfg.vPlaneN2PUpEn = 0x0;
vPlaneCfg.sWidth    = VIDEO_WIDTH;
vPlaneCfg.sHeight   = VIDEO_HEIGHT;
smtDMSetVCtrl(vPlaneCfg);
```

5.1.29. smtDMSetVBlndMode()/smtDMGetVBlndMode()

Description

DM(Display Module)의 video plane blend mode register를 read/write.

```
void smtDMSetVBlndMode(DMPlaneBlndMode blndMode);  
void smtDMGetVBlndMode(DMPlaneBlndMode *pBlndMode);
```

Parameters

blndMode

video plane blend mode register wrapper 구조체.

pBlndMode

video plane blend mode register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
vPlaneBlndMode.xPos = 0;  
vPlaneBlndMode.yPos = 0;  
vPlaneBlndMode.colKeyEn = 0x0;  
vPlaneBlndMode.blendMod = 0x0; //0:no alpha , 2: global alpha, 3: pixel alpha  
smtDMSetVBlndMode(vPlaneBlndMode);
```

5.1.30. smtDMSetVBlnd()/smtDMGetVBlnd()

Description

DM(Display Module)의 video plane blend register를 read/write.

```
void smtDMSetVBlnd(DMPlaneBlnd blnd);
void smtDMGetVBlnd(DMPlaneBlnd *pblnd);
```

Parameters

blnd

video plane blend register wrapper 구조체.

pBlnd

video plane blend register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
vPlaneBlnd.alpha = 0xFF;
vPlaneBlnd.colKey = 0xFFFFFFFF;
smtDMSetVBlnd(vPlaneBlnd);
```

5.1.31. smtDMSetVBase()/smtDMGetVBase()

Description

DM(Display Module)의 video plane base address register read/write.

```
void smtDMSetVBase(smtUint16 planeXRef);  
smtUint16 smtDMGetVBase(void);
```

Parameters

planeXRef

video plane XRef value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtDMSetVBase(0x0);
```

5.1.32. smtDMSetVAddr()/smtDMGetGVddr()

Description

DM(Display Module)의 video plane start address register를 read/write.

```
void smtDMSetVAddr(smtUInt32 addr);  
smtUInt32 smtDMGetVAddr(void);
```

Parameters

addr

video plane start addr value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtUInt16 *frame0 = (smtUInt16 *)VIDEO_BASEADDR;  
smtDMSetVAddr((smtUInt32)frame0);
```

5.1.33. smtDMSetVGamma()/smtDMGetVGamma()

Description

DM(Display Module)의 video plane gamma register를 read/write.

```
void smtDMSetVGamma(smtUInt32 addr, smtUInt32 gammaVal);  
smtUInt32 smtDMGetVGamma(smtUInt32 addr);
```

Parameters

addr

Gamma look up table index.

gammaVal

Gamma look up table index에 해당하는 gamma value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
none.
```

5.1.34. smtDMSetBGCol()/smtDMGetBGCol()

Description

DM(Display Module)의 back ground plane background color register를 read/write.

```
void smtDMSetBGCol(smtUInt32 colour);  
smtUInt32 smtDMGetBGCol(void);
```

Parameters

colour

background plane의 색상 값.(RGB888 단색만 지원)

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtDMSetBGCol(0xFF0000); //set back ground color to red
```

5.1.35. smtDMSetLCDCtrl()/smtDMGetLCDCtrl()

Description

DM(Display Module)의 Lcd control register를 read/write.

```
void smtDMSetLCDCtrl(DMLcdCtrl ctrl);
void smtDMGetLCDCtrl(DMLcdCtrl *pCtrl);
```

Parameters

ctrl

Lcd control register wrapper 구조체.

pCtrl

Lcd control register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMLcdCtrl dmLcdCtrl;

dmLcdCtrl.invertHSync      = LCD_IHS;
dmLcdCtrl.invertVSync     = LCD_IVS;
dmLcdCtrl.invertPixClk    = LCD_IPS;
dmLcdCtrl.invertWrEn      = LCD_IEO;
dmLcdCtrl.lcdBpp          = LCD_BPP;
dmLcdCtrl.rbSwap          = LCD_BGR;
dmLcdCtrl.pDiv             = 0x0;
dmLcdCtrl.lcdPwrEn        = 0x1;
dmLcdCtrl.lcdEn           = LCD_VIDEO_SYNC_EN; // video sync enable

smtDMSetLCDCtrl(dmLcdCtrl);
```

5.1.36. smtDMSetHSync0()/smtDMGetHSync0()

Description

DM(Display Module)의 lcd horizontal sync0 register를 read/write.

```
void smtDMSetHSync0(DMHSync0Ctrl hSync0);
void smtDMGetHSync0(DMHSync0Ctrl *pHSync0);
```

Parameters

hSync0

lcd hSync0 register wrapper 구조체.

pHSync0

lcd hSync0 register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMHSync0Ctrl dmHSync0;
dmHSync0.tHFrontPorch = LCD_HFP;
dmHSync0.tHBackPorch = LCD_HBP;
smtDMSetHSync0(dmHSync0);
```

5.1.37. smtDMSetHSync1()/smtDMGetHSync1()

Description

DM(Display Module)의 lcd horizontal sync1 register를 read/write.

```
void smtDMSetHSync1(DMHSync0Ctrl hSync1);
void smtDMGetHSync1(DMHSync0Ctrl *pHSync1);
```

Parameters

hSync1

lcd hSync1 register wrapper 구조체.

pHSync1

lcd hSync1 register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMHSync1Ctrl dmHSync1;
dmHSync1.tHDspPixPerLine = LCD_CPL;
dmHSync1.tHPulsWidth = LCD_HSW;
smtDMSetHSync1(dmHSync1);
```

5.1.38. smtDMSetVSync0()/smtDMGetVSync0()

Description

DM(Display Module)의 lcd vertical sync0 register를 read/write.

```
void smtDMSetVSync0(DMVSyn0Ctrl vSync0);  
void smtDMGetVSync0(DMVSyn0Ctrl *pVSync0);
```

Parameters

vSync0

lcd vSync0 register wrapper 구조체.

pVSync0

lcd vSync0 register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMVSyn0Ctrl dmVSync0;  
  
dmVSync0.tvFrontPorch    = LCD_VFP;  
dmVSync0.tvBackPorch    = LCD_VBP;  
smtDMSetVSync0(dmVSync0);
```

5.1.39. smtDMSetVSync1()/smtDMGetVSync1()

Description

DM(Display Module)의 lcd vertical sync1 register를 read/write.

```
void smtDMSetVSync1(DMVSyn1Ctrl vSync1);  
void smtDMGetVSync1(DMVSyn1Ctrl *pVSync1);
```

Parameters

vSync1

lcd vSync1 register wrapper 구조체.

pVSync1

lcd vSync1 register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMVSyn1Ctrl dmVSync1;  
  
dmVSync1.tVDspPeriod   = LCD_LPS;  
dmVSync1.tVPulsWidth   = LCD_VSW;  
smtDMSetVSync1(dmVSync1);
```

5.2. VIDEO ENCODER

5.2.1. smtSetVideoEncode()

Description

set video encoder의 control / internal control registers.

```
void  
smtSetVideoEncode(  
    VEncCtrl *vCtrl,  
    VEncInternal *vInternal  
);
```

Parameters

vCtrl

video encoder control register 정보를 담고 있는 구조체의 포인터.

vInternal

video encoder internal control register 정보를 담고 있는 구조체의 포인터.

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: videoenc_pre_drv.h

Source: videoenc_pre_drv.c

Library: none

Example Code

```
VEncInternal vInternal;  
vInternal.hSyncWidth    = HSYNC_WID;  
vInternal.burstWidth    = BURST_WID;  
vInternal.lDelay        = LUMA_DELAY;  
vInternal.cDelay        = CHRO_DELAY;  
vInternal.patternMode   = EN_INTERNAL_PATTERN;  
vInternal.pattern       = COLOR_PATTERN_MODE;  
vInternal.resetSCH      = EN_REST_SCH;  
vInternal.color         = EN_COLOR_KILL;  
vInternal.CFilter       = CHRO_FILTER_SEL;  
vInternal.lFilter       = LUMA_FILTER_SEL;  
  
smtSetVideoEncode(&vCtrl,&vInternal);
```

5.2.2. smtSetVideoStatus()/smtGetVideoStatus()

Description

set/get status register of video encoder.

```
void  
smtSetVideoStatus(  
    VEncStatus *vStatus  
);  
void smtGetVideoStatus(  
    VEncStatus *vStatus  
);
```

Parameters

vStatus

video encoder status register 정보를 담고 있는 구조체의 포인터.

Return Value

none

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: videoenc_pre_drv.h

Source: videoenc_pre_drv.c

Library: none

Example Code

```
VEncStatus vStatus;  
  
//set Video Encoder Status Register  
vStatus.Enable          = 0x1; // enable video encoder  
vStatus.fieldCount      = 0x0;  
vStatus.Hcount          = 0x0;  
vStatus.Vcount          = 0x0;  
smtSetVideoStatus(&vStatus);
```

5.2.3. smtSetVideoSubCarrier()

Description

set the subcarrier frequency setup / subcarrier adfust phase register of video encoder.

```
void smtSetVideoSubCarrier(  
    smtUInt16 subPhase,  
    smtUInt32 subReq  
);
```

Parameters

subPhase

subcarrier phase value.

subReq

subcarrier frequency setting value.

Return Value

none

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: videoenc_pre_drv.h

Source: videoenc_pre_drv.c

Library: none

Example Code

None.

5.3. VIF

5.3.1. smtVIFSetMode()/smtVIFGetMode()

Description

set/get control register of external video input.

```
void smtVIFSetMode (VIF_CTRL_STRUCT vif);
void smtVIFGetMode (VIF_CTRL_STRUCT vif);
```

Parameters

vif

VIF control control register 정보를 담고 있는 구조체

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: vif_pre_drv.h

Source: vif_pre_drv.c

Library: none

Example Code

```
vif.hBlank      = 0;
vif.vBlank      = 0;
vif.field       = 0;
vif.ycOrder     = 2;           // y/cb/y/cr order
vif.startIntEn  = 1;           // Frame start interrupt enable
vif.endIntEn    = 0;           // Frame end interrupt enable
vif.dmaEn       = 1;           // DMA enable

smtVIFSetMode(vif);
```

5.3.2. smtVIFGetIntStatus()

Description

set/get status register of video encoder.

```
smtUInt8 smtVIFGetIntStatus (void);
```

Parameters

none

Return Value

VIF start/end frame flag

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: vif_pre_drv.h

Source: vif_pre_drv.c

Library: none

Example Code

```
smtUInt8 vifStatus;  
vifStatus = smtVIFGetIntStatus();
```

5.3.3. smtVIFSetProperty()/smtVIFGetProperty()

Description

set/get control register of external video input.

```
void smtVIFSetProperty (VIF_PROP_STRUCT vifProp);  
void smtVIFGetProperty (VIF_PROP_STRUCT vifProp);
```

Parameters

vifProp

vif property(position, size, address) 정보를 담고 있는 구조체.

Return Value

none

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: vif_pre_drv.h

Source: vif_pre_drv.c

Library: none

Example Code

```
vifProp.vifDmaAddr= FRAME_BASEADDR;  
vifProp.vifXPos   = 0;  
vifProp.vifYPos   = 0;  
vifProp.vifXSize  = 720;  
vifProp.vifYSize  = 240;  
  
smtVIFSetProperty(vifProp);
```

6. STORAGE DEVICES

6.1.1. NAND_OPT0

Description

NAND 1st Operation register bit wrapper

```
typedef struct
{
    smtBoolean    chipSel;
    smtUInt8      option;
    smtUInt16     dataSize;
    smtUInt8      opMode;
    smtUInt8      TransSize;
    smtUInt8      cmdAddrTransByte;
    smtUInt8      cmdAddrFlag;
} NAND_OPT0;
```

Parameters

chipSel

NAND chip select bit

option

Select operation option

VALUE	Description
0	No operation
1	Ready and Busy wait
2	Auto read status
3	Continue next command

dataSize

Read/write size in page, If IO width 8bits then operation unit is byte, if IO width 16bits Then operation unit is half word, so on..

opMode

Select operation mode

VALUE	Description
0	Read data operation
1	Read status operation
2	Read ID operation
3	Write data operation
4	Other case is No operation

TransSize

FIFO transfer size, if IO width is 16bits then not support byte tranfer

VALUE	Description
0	Byte transfer
1	Half word transfer
2	Word transfer

cmdAddrTransByte

NAND command byte and address byte (command byte + address byte)

cmdAddrFlag

NAND command or address flag bit, LSB means first cycle of NAND command or address

VALUE	Description
0	Address flag
1	Command flag

Return value

None

Remarks

None

Requirements

Header: nand_pre_drv.h

Source: none

Library: none

6.1.2. NAND_OPT1

Description

NAND 2nd Operation register bit wrapper

```
typedef struct
{
    smtUInt8  CMDADDR[4];
} NAND_OPT1;
```

Parameters

CMDADDR[0~3]

NAND address command byte

FIELD	Description
CMDADDR[0]	1 st command addrss cycle
CMDADDR[1]	2 nd command addrss cycle
CMDADDR[2]	3 rd command addrss cycle
CMDADDR[3]	4 th command addrss cycle

Return value

None

Remarks

None

Requirements

Header: nand_pre_drv.h

Source: none

Library: none

6.1.3. NAND_OPT2

Description

NAND 3rd Operation register bit wrapper

```
typedef struct
{
    smtUInt8  CMDADDR[4];
} NAND_OPT2;
```

Parameters

CMDADDR[0~3]

NAND address command byte

FIELD	Description
CMDADDR[0]	5 th command addrss cycle
CMDADDR[1]	6 th command addrss cycle
CMDADDR[2]	7 th command addrss cycle
CMDADDR[3]	8 th command addrss cycle

Return value

None

Remarks

None

Requirements

Header: nand_pre_drv.h

Source: none

Library: none

6.1.4. NAND_CFG

Description

NAND configuration register bit wrapper

```
typedef struct
{
    smtBoolean      nandBootEn;
    smtBoolean      IOWidth;
    smtBoolean      NandWidth;
    smtUInt8        bootCfg;
    smtBoolean      OutDtmn;
    smtUInt8        TADLTWB;
    smtUInt8        TCALS;
    smtUInt8        TRWLP;
    smtUInt8        TRWHP;
} NAND_CFG;
```

Parameters

nandBootEn

NAND boot enable bits

IOWidth

NAND IO width bits

VALUE	Description
0	Enable 8bits I/O BUS
1	Enable 16bits I/O BUS

NandWidth

NAND type bits

VALUE	Description
0	Enable 8bits NAND cell
1	Enable 16bits NAND cell

bootCfg

NAND address type

VALUE	Description
0	3 address cycle NAND (1 column , 2 row address)
1	4 address cycle NAND (1 column, 3 row address)
2	4 address cycle NAND (2 column, 2 row address)
3	5 address cycle NAND (2 column, 3 row address)

OutDtmn

Enable twin NAND operation, if IO width 16bits and attach two 8bit NAND to NAND Controller then send same command to two 8bit NAND

TADLTWB

NAND configuration register bit of CT500 manual

TCALS

NAND configuration register bit of CT500 manual

TRWLP

NAND configuration register bit of CT500 manual

TRWHP

NAND configuration register bit of CT500 manual

Return value

None

Remarks

None

Requirements

Header: nand_pre_drv.h

Source: none

Library: none

Confidential

6.1.5. NAND_CTRL

Description

NAND control register bit wrapper

```
typedef struct
{
    smtBoolean    NFCtrlRst;
    smtUInt8      FIFOLevel;
    smtBoolean    Ecc512byteEn;
    smtBoolean    autoEccWr;
    smtBoolean    DMAEn;
    smtBoolean    wrEndIntEn;
    smtBoolean    rdEndIntEn;
    smtBoolean    FIFOIntEn;
    smtBoolean    RnBintEn1;
    smtBoolean    RnBintEn0;
} NAND_CTRL;
```

Parameters

NFCtrlRst

NAND controller reset bit

FIFOLevel

NAND FIFO level setting value

NOTICE: You should refer to NAND test code (nand.c), because FIFO level relative to NAND transfer size and DMA transfer size

Ecc512bteEn

512byte ECC enable bits

autoEccWr

Auto write ECC enable bits. ECC write position refer to CT500 manual NAND session

DMAEn

NAND DMA enable bits

wrEndIntEn

Write available interrupt enable bits, avail means ready to write to FIFO as user

FIFO level

rdEndIntEn

Read available interrupt enable bits, avail means ready to read to FIFO as user

FIFO level

FIFOIntEn

FIFO interrupt

RnBintEn1

Ready and busy interrupt enable bits for NAND bank 1

RnBintEn0

Ready and busy interrupt enable bits for NAND bank 0

Return value

None

Remarks

None

Requirements

Header: nand_pre_drv.h

Source: none

Library: none

Confidential

6.1.6. NAND_STAT

Description

NAND status register bits wrapper

```
typedef struct
{
    smtBoolean  NFCtrlBusy;
    smtBoolean  NFIDValid;
    smtBoolean  NFStatValid;
    smtUInt16   NFStatus;
    smtBoolean  wrEnd;
    smtBoolean  rdEnd;
    smtBoolean  wrFIFOReady;
    smtBoolean  rdFIFOReady;
    smtBoolean  RnBDetect0;
    smtBoolean  RnBSTAT0;
} NAND_STAT;
```

Parameters

NFCtrlBusy

NAND controller busy status.

NFIDValid

NAND ID valid status

NFStatus

NAND cell status, refer to data sheet

wrEnd

Write operation end status. It's occurred, if NAND controller send user transfer size all

rdEnd

Read operation end status. It's occurred, if NAND controller receive user transfer size all

wrFIFOReady

Write available status, avail means ready to write to FIFO as user FIFO level

rdFIFOReady

Read available status, avail means ready to read to FIFO as user FIFO level

RnBDetect0

Ready and busy detected status

RnBSTAT0

Bank 1 NAND cell Ready and busy pin mapping bits

Return value

None

Remarks

None

Requirements

Header: nand_pre_drv.h

Source: none

Library: none

Confidential

6.1.7. NAND_FIFOSTAT

Description

NAND FIFO status register bits wrapper

```
typedef struct
{
    smtUInt8    DFIFOLevel1;
    smtUInt8    DFIFOLevel0;
    smtUInt8    QLevel;
}    NAND_FIFOSTAT;
```

Parameters

DFIFOLevel0/1

NAND FIFO level

Return value

None

Remarks

None

Requirements

Header: `nand_pre_drv.h`

Source: none

Library: none

6.1.8. NAND_ECC

Description

NAND main/spare ECC register bits wrapper

```
typedef struct
{
    smtUint32    ecc[16];
    smtUint32    secc[16];
} NAND_ECC;
```

Parameters

[ecc\[0~15\]](#)

NAND main ECC register for 0 ~ 15 sectors

[secc\[0~15\]](#)

NAND spare ECC register for 0 ~ 15 sectors

Return value

None

Remarks

None

Requirements

Header: nand_pre_drv.h

Source: none

Library: none

6.1.9. smtNANDSetOPT0()

Description

NAND 1st Operation's software interface

```
void smtNANDSetOPT0(NAND_OPT0 *popt0);
```

Parameters

popt0

NAND 1st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

Refer to test code (peri\storage\nand.c)

6.1.10. smtNANDSetOPT1()

Description

NAND 2st Operation's software interface

```
void smtNANDSetOPT1(NAND_OPT1 *popt1);
```

Parameters

popt1

NAND 1st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

Refer to test code (peri\storage\nand.c)

6.1.11. smtNANDSetOPT2()

Description

NAND 3st Operation's software interface

```
void smtNANDSetOPT1(NAND_OPT2 *popt2);
```

Parameters

popt1

NAND 2st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

Refer to test code (peri\storage\nand.c)

6.1.12. smtNANDGetData()/smtNANDSetData()

Description

NAND, NFDATA register software interface

```
void smtNANDSetData(smtUInt32 *pdata)  
smtUInt32 smtNANDGetData(void)
```

Parameters

pdata

data for write NAND

Return Value

Data for read from NAND.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

Refer to test code (peri\storage\nand.c)

6.1.13. smtNANDSetCfg()/smtNANDGetCfg()

Description

NAND, NCONF register software interface

```
void smtNANDSetCfg(NAND_CFG *pconfig)
void smtNANDGetCfg(NAND_CFG *pconfig)
```

Parameters

pconfig

NAND, NCONF register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

Refer to test code (peri\storage\nand.c)

6.1.14. smtNANDSetControl()/smtNANDGetControl()

Description

NAND, NFCTRL register software interface

```
void smtNANDSetControl(NAND_CTRL *pcontrol)
void smtNANDGetControl(NAND_CTRL *pcontrol)
```

Parameters

pcontrol

NAND, NFCTRL register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

Refer to test code (peri\storage\nand.c)

6.1.15. smtNANDSetStatus()/smtNANDGetStatus()

Description

NAND, NFSTAT register software interface

```
void smtNANDSetStatus(NAND_STAT *pstatus)  
void smtNANDGetStatus(NAND_STAT *pstatus)
```

Parameters

pstatus

NAND, NFSTAT register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

Refer to test code (peri\storage\nand.c)

6.1.16. smtNANDGetFIFOStatus()

Description

NAND, NFFIFOSTAT register software interface

```
void smtNANDGetFIFOStatus(NAND_FIFOSTAT *pfifoStatus)
```

Parameters

pstatus

NAND, NFFIFOSTAT register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.17. smtNANDGetECC()

Description

NAND, ECCSECTOR0~ECCSECTOR8, SECCSECTOR0~SECCSECTOR3 register software interface

```
void smtNANDGetECC(smtUint8 eccIdx, smtUint32 *pecc, smtUint32 *psecc)
```

Parameters

eccIdx

wanted Main/Spare ECC number

pecc

smtUint32 pointer for Main ECC, if NULL no operation for this pointer

psecc

smtUint32 pointer for Spare ECC, if NULL no operation for this pointer

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand_pre_drv.c

Library: none

Example Code

Refer to test code (peri\storage\nand.c)

6.2. PSRAM

6.2.1. smtPSRAMInit()

Description

PSRAMCON, PSRAMTCON, PSRAMTOUT register control function.

```
void smtPSRAMInit (smtUInt32 ToutValue)
```

Parameters

ToutValue

Timeout Value

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: psram_pre_drv.h

Source: psram_pre_drv.c

Library: none

Example Code

```
none
```

6.3. SRAM

6.3.1. smtSMCGetControl()/smtSMCSetControl()

Description

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register software interface

```
void smtSMCGetControl(smtUInt8 bank, SMC_CTRL *pcontrol)
void smtSMCSetControl(smtUInt8 bank, SMC_CTRL *pcontrol)
```

Parameters

bank

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register selector, i.e. 0 means ESMC_B0_CON.

pcontrol

ESMC_B[0-3]_CON register's 1:1 mapping layer

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: smc_pre_drv.h

Source: smc_pre_drv.c

Library: none

Example Code

```
none
```

6.3.2. smtSMCGetControl()/smtSMCSetControl()

Description

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register software interface

```
void smtSMCGetControl(smtUInt8 bank, SMC_CTRL *pcontrol)
void smtSMCSetControl(smtUInt8 bank, SMC_CTRL *pcontrol)
```

Parameters

bank

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register selector, i.e. 0 means ESMC_B0_CON.

pcontrol

ESMC_B[0-3]_CON register's 1:1 mapping layer

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: smc_pre_drv.h

Source: smc_pre_drv.c

Library: none

Example Code

```
none
```

6.4. DDR-SDRAM

6.4.1. smtDMCSetControl()/smtDMCGetControl()

Description

DDRCON register software interface

```
void smtDMCSetControl(DMC_CTRL *pcontrol)
void smtDMCGetControl(DMC_CTRL *pcontrol)
```

Parameters

pcontrol

DDRCON register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

6.4.2. smtDMCGetTimingControl()/smtDMCSetTimingControl()

Description

DDRTCON register software interface

```
void smtDMCSetTimingControl(DMC_TIMING *ptiming)
void smtDMCGetTimingControl(DMC_TIMING *ptiming)
```

Parameters

ptiming

DDRTCON register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

6.4.3. smtDMCGetPowerControl()/smtDMCSetPowerControl()

Description

DDRPCON register software interface

```
void smtDMCSetPowerControl(DMC_POWER *ppower)  
void smtDMCGetPowerControl(DMC_POWER *ppower)
```

Parameters

ppower

DDRPCON register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

6.4.4. smtDMCGetRefreshControl()/smtDMCSetRefreshControl()

Description

DDRREF register software interface

```
void smtDMCSetRefreshControl(DMC_REFRESH *prefresh)
void smtDMCGetRefreshControl(DMC_REFRESH *prefresh)
```

Parameters

prefresh

DDRREF register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```